

Statistical mechanical characterization of memory and quench response in state-separated LLM-agent networks

Alexander V. Mantzaris

School of Data, Mathematical, and Statistics Sciences

University of Central Florida

Orlando, FL 32816, USA

`alexander.mantzaris@ucf.edu`

Abstract

Populations of interacting large-language-model (LLM) agents can develop collective dynamics that are not visible in individual responses or final averages. Statistical mechanics provides a principled framework for measuring these dynamics through order, fluctuations, correlations, response, persistence, and temporal asymmetry. We treat networks of locally informed LLM agents as finite interacting stochastic systems with private state, bounded memory, and networked communication.

We study modular networks of 16 Qwen and Granite agents under random sequential updates. The study combines a corrected analysis of an earlier experiment, without altering its trajectories, with a prospectively specified cross-model experiment comprising 48 trajectories, 34,560 local decisions, and six independent graph environment clusters per model. We reverse and restore an external field while comparing perturbed systems with matched nominal trajectories. To isolate memory, we compare persistent histories with current state Markovized prompts and format-matched scrambled histories.

All four prespecified tests met their inferential criteria. Field reversal produced a cross model response of 42.3 within model distance units (95% CI [24.3, 59.3]), and the fixed restoration contrast was 52.5 units (95% CI [35.4, 68.7]). Persistent memory increased bias adjusted and path reversal divergence by 0.054 nats per attempted update relative to Markovized histories (95% CI [0.035, 0.075]) and by 0.048 relative to scrambled histories (95% CI [0.022, 0.075]). The latter comparison indicates that temporally meaningful history matters beyond the presence of additional prompt content, with especially pronounced and consistent effects for Granite. An out of sample kinetic surrogate identified which aspects of the response admit a simple local closure and which remain model and history dependent.

These results show that statistical mechanical observables recover collective response, memory effects, restoration dynamics, and reduced structure that population means discard. The interpretation remains effective and finite size decoding temperature is not physical temperature, reference energy is not literal energy, and projected path reversal divergence is not exact thermodynamic entropy production.

Keywords: statistical mechanics; agents; language-model agents; quench dynamics; entropy rate; multi-information; temporal asymmetry.

1 Introduction

Large language models are increasingly studied not only as isolated prompt response systems but as components of interacting agent populations [1, 2]. Once agents exchange messages, retain histories, and act through a network, an output from one local decision can become part of another agent’s next input. This feedback permits amplification, cascades, coordination, fragmentation, lock-in, delayed response, and incomplete recovery even when every component uses fixed model weights. The capability of one model instance therefore does not determine the trajectory of the group, because repeated circulation can create modes of organization that no single generation exhibits. The scientific question becomes what dynamical organization emerges when many stochastic language-model instances interact over time, a question that matters for the analysis, design, reliability, and control of multi-agent AI systems.

At the component level, an LLM agent places a generative model inside a recurrent loop of observation, decision, action, and environmental feedback. Its future inputs can contain its own earlier outputs, received messages, and changes caused by earlier actions. Population trajectories can consequently depend on message content, network position, update order, retained context, and decoding randomness. Conventional benchmarks can characterize useful local capabilities, but they do not reveal whether network feedback produces stable organization, amplifies a perturbation, stores hidden history, or returns the population to its earlier regime. Those are properties of the interacting process rather than of any response considered alone.

Recent agent architectures establish both the feasibility of such processes and the source of their complexity. Role-playing agents cooperate through natural-language exchange [3], while generative agents combine observation, planning, reflection, and persistent natural-language memory in a simulated social environment [4]. These systems show that retrieved history and communication can materially condition later actions. Yet architecture and application studies are commonly evaluated by task completion or behavioral plausibility. Those objectives do not by themselves specify a controlled stochastic process, distinguish private state from observed state, or identify which population coordinates retain dynamical information.

Computational social simulation and population experiments make the same measurement problem visible from another direction. Conditioned language models reproduce portions of survey-response distributions [5], and simulated samples replicate some established behavioral experiments while exhibiting model specific distortions, in [6] the broader opportunities and validity limits of LLM-empowered agent-based simulation are reviewed in [7]. Networked agents have generated consensus and fragmentation in opinion experiments [8], shared naming conventions and collective bias [9], and model-dependent cooperation and coordination in repeated games [10]. Together, these studies establish that interaction can produce reproducible group level outcomes, while providing lessons against treating semantic possibility as a dynamical explanation or an LLM population as a faithful proxy for human behavior.

When agents interact through a network, their collective behavior depends not only on how individual agents update, but also on which agents can communicate with one another. The interaction graph defines the routes along which information, influence, and external perturbations can spread. A densely connected network may transmit a local change throughout the population, whereas a modular network may initially confine that change to a single community. Statistical mechanics and network science provide tools for relating these microscopic interaction patterns to macroscopic collective behavior [11, 12]. Previous

studies of cooperative processes on networks have also shown that network topology and finite population size can substantially affect the collective behavior that is observed [13]. Population averages alone cannot reveal this internal organization. Two agent populations can have the same mean level of order while differing in their fluctuations, correlations, community structure, and response to perturbation. We therefore measure how strongly agent states are correlated as a function of their distance on the interaction graph. These graph-distance correlations indicate whether collective organization remains localized among neighboring agents or extends across the wider network. Such network-aware observables complement population means and provide a more complete description of how interacting agent groups organize and respond.

More broadly, statistical mechanics supplies a disciplined framework for connecting microscopic interactions to macroscopic behavior. Agent-based models ask how local rules and heterogeneous interactions generate system-level outcomes [14], while statistical physics has developed quantitative descriptions of opinion, cultural, language, and consensus dynamics [15, 16]. In a conventional stochastic model, the microscopic rule may be simple: a spin flips with a specified probability, or an agent adopts the state of a neighbor. Such local rules can nevertheless generate collective relaxation [17], stationary behavior constrained by a potential [18], or distinct forms of organization when interactions are nonreciprocal [19].

An LLM agent makes the same micro-to-macro question more challenging because its local update is not described by a simple flip probability. When selected for an update, an agent interprets a language prompt containing its current state, messages from neighboring agents, an external field, and, when available, a private interaction history. For example, an agent receiving several neighboring messages in favor of one action may follow that local majority, resist it because of its remembered history, or produce a different choice under another stochastic generation. The microscopic transition is therefore language-conditioned, history-dependent, and high-dimensional, even when the recorded belief and action are categorical. Statistical-mechanical observables make the resulting population dynamics tractable without requiring a complete analytic model of every language-mediated decision. They allow us to ask whether local updates create population order, whether perturbations produce a coordinated response, whether correlations extend across the network, and whether private memory changes the apparent temporal dynamics.

For a finite population observed over finite time windows, fluctuations, correlations, response distributions, and variation across independent realizations contain information that a mean trajectory discards. Statistical mechanics can therefore be used here as a measurement framework for collective LLM-agent behavior without assuming literal particles, a thermodynamic limit, or the existence of a phase transition.

Controlled perturbation is important because nominal or steady observations underdetermine dynamics. Similar baseline averages could describe a robustly organized system, a fragile one, a system that responds strongly and recovers, or one with delayed or incomplete recovery. Reversing an external field and then restoring it probes finite-horizon sensitivity, response, relaxation, and recovery directly. Comparing that trajectory with a matched nominal arm also separates intervention-associated change from ordinary evolution over the same observation window. A quench is used here as a dynamical intervention, not as evidence that the network is an equilibrium material.

Persistent history creates a distinct coarse-graining problem. The complete agent process may include private textual state that is absent from a measured belief then action projection, so the observed population trajectory need not be Markov even when the augmented simulator state is. Comparing persistent and Markovized prompts tests the effect of retaining history; comparing genuine history with a scrambled, format-matched

control helps separate temporally related information from the mere presence of additional prompt material. Temporal correlation then measures persistence, while path reversal tests ordering in the projected trajectory. State-space currents and trajectory reversal have exact thermodynamic interpretations in appropriate fully observed Markov systems [20, 21], but coarse graining can hide dissipation and make inferred bounds estimator dependent [22, 23]. The path statistic used here is therefore projected temporal asymmetry, not literal entropy production.

Recent LLM-population studies approach these questions with statistical-physics concepts, group size can alter collective misalignment [24], topology and self social weighting can change conformity [25], intrinsic fields can outweigh fitted neighbor coupling [26], debate noise can produce model dependent finite size crossovers [27], and high dimensional interaction data can admit compact dynamical regimes [28]. This progress narrows the gap, but the joint problem remains insufficiently resolved that controlled state and information boundaries, field reversal and restoration, genuine and placebo history interventions, spatial and temporal diagnostics, cluster-level inference, and an out-of-sample closure test are rarely examined together. A task score or single consensus coordinate cannot answer all of these questions.

No single measurement is sufficient to describe how a population of LLM agents organizes. For example, two populations can have the same average belief while differing substantially in their internal behavior. One may be relatively uniform, while the other may contain distinct communities or fluctuate between different collective states. We therefore use several complementary measurements. Belief and action order quantify the extent of population alignment, while their overlap indicates whether the agents’ stated beliefs are reflected in their committed actions. Fluctuations, entropy, dependence, and effective compatibility then show whether this alignment is stable, variable, coordinated, or structured by interactions among neighboring agents. Graph-distance correlations reveal whether changes remain local or extend across the network. Autocorrelation and Binder statistics provide additional finite-window descriptions of persistence and of the distribution of collective order. These diagnostics characterize the observed trajectories but are not treated as evidence of equilibrium correlation times, critical slowing down, or a phase transition.

The experimental controls address a different problem: determining what causes the observed collective differences. A field quench is compared with matched nominal evolution so that ordinary changes over time are not mistaken for a response to the perturbation. Persistent memory is compared with both a memory-free condition and a scrambled-history condition, allowing us to distinguish the effect of meaningful past interactions from the mere addition of more prompt content. Coarse-grained block reversal measures how these different history conditions alter the temporal ordering of the population trajectory. Statistical inference is performed over independent graph–environment clusters so that repeated updates within the same network are not incorrectly treated as independent evidence. Finally, an out-of-sample kinetic surrogate tests whether the observed collective response can be summarized by a small number of macroscopic variables. Success would support a useful reduced description, while failure identifies information that those macroscopic variables omit. Together, these measurements and controls prevent a change in a population average from being mistaken for a complete account of collective organization, memory, and response.

The contribution is not a general-purpose agent framework, a performance benchmark, or a model of human society. It is a controlled finite-size study that treats interacting LLM instances as measurable stochastic components and examines their collective response to external fields, retained and control histories, network organization, and coarse-graining.

The direct experiments use two pinned instruction-model families, $N = 16$ reciprocal modular networks, and finite observation windows; the interpretation is effective and model-specific, with no thermodynamic-limit, exact physical entropy-production, or universality claim. Section 2 defines the network and interventions, Section 3 defines measurement and inference, and Section 4 sets out the study roles. Sections 5–6 present the quench and memory evidence, Section 7 tests reduced closure, and Sections 8 & 9 give the interpretation and conclusions.

2 State-separated LLM-agent networks and quench protocols

2.1 Network state and update dynamics

For agent i at attempted update t , the primary categorical coordinates are belief $b_i(t) \in -1, +1$ and committed action $a_i(t) \in -1, +1$. From a statistical-mechanical perspective, these variables play a role similar to spin coordinates, but they represent two distinct aspects of the agent. Belief records the agent’s stated position, whereas action records the choice that it commits to the interacting system. Keeping them separate allows the model to represent cases in which an agent’s expressed belief and externally consequential action do not coincide.

An LLM agent also has a richer local state than a conventional spin. Its bounded state includes confidence, commitment status, workload, a memory code, at most three private history entries, an inbox, and an outbox. These variables supply context for the next language-conditioned update. Consequently, two agents with the same current belief and action can respond differently because they have received different messages, retain different private histories, or have different commitment states. The microscopic transition rule is therefore not determined by the visible binary coordinates alone.

Display labels are counterbalanced across experimental conditions and decoded back to latent spins. This prevents a particular word, label, or display position from being permanently associated with either -1 or $+1$. For statistical analysis, the selected microscopic coordinates are collected in the projection

$$Y_t = (\mathbf{b}_t, \mathbf{a}_t, \mathbf{c}_t, \mathbf{m}_t). \quad (1)$$

This projection provides a tractable representation of the interacting population, but it is not the complete simulator state. In particular, private history and message context can influence future updates even when two projected states appear identical. Distinguishing the observed projection from the complete state is therefore essential when interpreting memory, Markovianity, and temporal asymmetry in the resulting population dynamics.

Let

$$\Xi_t = (S_{1,t}, \dots, S_{N,t}, G_t, \mathcal{D} * t, q_t, R_t) \quad (2)$$

denote all agent-private states $S * i, t$, graph state G_t , delivery operator $\mathcal{D}_t$, quench phase q_t , and the explicitly specified randomness state R_t . The quench phase records whether the system is in its initial field condition, the reversed-field condition, or the subsequent restored-field condition. Reversing the field supplies a controlled external perturbation, allowing us to measure how strongly the agent population responds, while restoring it allows us to examine whether and how rapidly the population returns toward its earlier behavior.

The scheduler is also part of the dynamical specification because it determines which agent receives an update opportunity at each attempted update, although it does not determine the action that the selected agent produces. Under random-sequential updating, the order of these opportunities affects which messages and neighbor states are available when later agents respond. Thus, even with the same initial state and quench schedule, different scheduler realizations can produce different microscopic trajectories. With fixed model weights and tokenizer, reconstructed local prompts, and a defined scheduler, Ξ_t induces a time-inhomogeneous transition kernel because the external conditions change across the quench phases. The evaluator’s map $Y_t = \phi(\Xi_t)$ deliberately omits some prompt history and randomness. Consequently, Y_t and any rolling macrostate need not be Markov.

Figure 1 summarizes the architecture that connects local LLM-agent interactions to the statistical-mechanical analysis. It first distinguishes the information available to an individual agent from the additional graph, scheduling, delivery, and randomness state maintained by the simulator. This boundary prevents the complete simulator state from being mistaken for information that any agent can observe or use. The figure also shows that the scheduler only selects an agent and delivers its available packet, whereas the LLM produces the resulting belief, action, memory update, and outgoing signal. Finally, it follows the successive reduction from the complete augmented state Ξ_t , to the evaluator’s microscopic projection Y_t , and then to the rolling macrostate Z_t . This map makes explicit where private history and microscopic detail are omitted, providing the conceptual basis for the later analysis of quench response, memory, and apparent non-Markovian dynamics.

Augmented process Ξ_t and measured projections

$$Y_t = \phi(\Xi_t); Z_t = \psi(Y_{t-w+1:t}) \text{ (evaluator-side)}$$

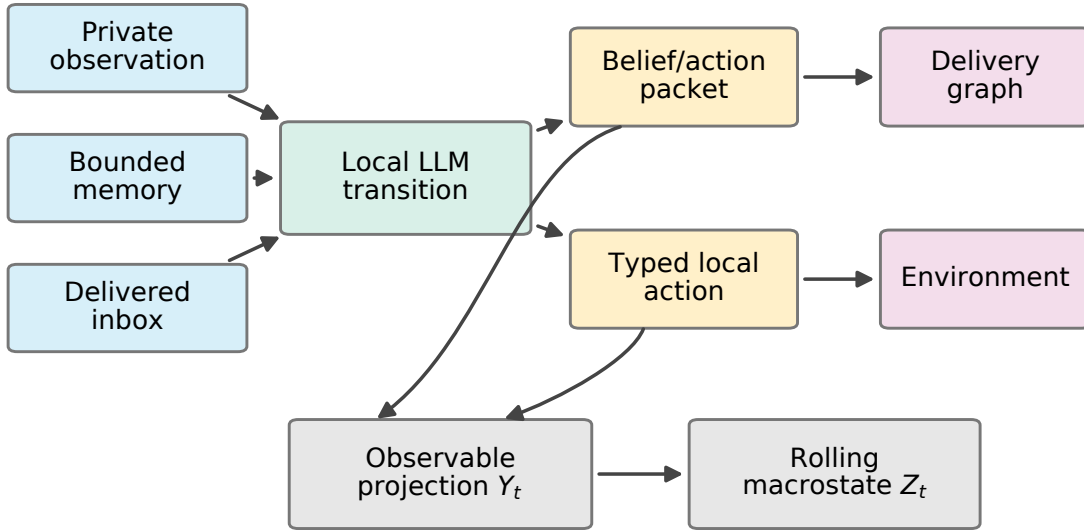


Figure 1: Agent information boundaries and the augmented-to-macroscopic map. The complete state Ξ_t includes private memory, graph and schedule state; the evaluator observes $Y_t = \phi(\Xi_t)$ and constructs rolling $Z_t = \psi(Y_{t-w+1:t})$. The scheduler offers a turn and delivers a packet but does not supply the model’s belief, action, memory, or signal.

At each attempted update, a random permutation gives every agent exactly one opportunity per sweep. Only the scheduled instance receives its private observation, current

local state, currently delivered inbox, and, in a memory condition, its own bounded history. It returns a validated typed object containing belief, action, confidence, commitment, memory state, outgoing signal, and tool action. One bounded repair may correct syntax; an invalid result causes no centrally selected scientific action. A serialized packet is sent on one graph edge. Fingerprints of nonrecipient peer-private states are compared around every transition. The corresponding information-boundary and implementation checks are described in Appendix D.

2.2 History conditions and matched controls

The agents perform a repeated coordination task between two counterbalanced symbolic alternatives, **amber** and **cobalt**. Neither alternative is intrinsically preferred, and the agents are not explicitly instructed to reach consensus. At each update opportunity, the selected agent receives a moderate private observation favoring one alternative, its current local state, and any packets recently delivered by its graph neighbors. It then returns a structured decision specifying its belief, committed action, confidence, commitment and memory states, and an outgoing signal.

Communication is therefore a form of bounded local signaling rather than free-form conversation. A neighbor packet reports the sender’s current belief, action, outgoing signal, confidence, commitment status, and bounded-memory category. In practical terms, it tells the receiving agent which alternative the sender currently favors, what it acted upon, and how confidently or firmly it holds that position. Agents do not receive their neighbors’ private observations, complete prompts, personal histories, or undelivered messages, and they never observe the global network state. The Markovized prompt is reconstructed from the explicitly recorded current state and currently available packets, without carrying free conversational history. The persistent prompt contains the same information but adds the selected agent’s last three own entries in the fixed format **time**, **belief**, **action**, **memory**. Thus, two trajectories with identical visible $(\mathbf{b}_t, \mathbf{a}_t)$ can have different conditional future laws because their agents may remember different earlier decisions:

$$P(Y_{t+1} \mid Y_t, M_t) \neq P(Y_{t+1} \mid Y_t, M'_t). \quad (3)$$

If M_t is projected out, conditional dependence on older visible states can remain. This provides a concrete mechanism for increased block-reversal divergence, but it is not a claim of literal physical dissipation.

The scrambled-history control tests whether an apparent memory effect could instead arise from the mere presence and length of an additional prompt section. It retains the same agent’s past opportunity times and uses the same number of entries, field names, and format, while the belief, action, and memory content is deterministically randomized from a prespecified control seed. No donor agent, future time, or hidden environmental outcome is used. Token counts and their paired differences are measured, although exact turn by turn equality is not assumed. Prompt balance diagnostics are shown in Figure 12 of Appendix B.

2.3 Field quench and restoration protocol

Each trajectory contains 15 baseline sweeps, 15 intervention sweeps, and 15 restoration sweeps. One sweep consists of $N = 16$ attempted updates, giving each agent one update opportunity on average under random-sequential scheduling. The baseline allows collective behavior to develop under the initial fields; the intervention permits the reversed evidence

to propagate through repeated local interactions; and the equally long restoration phase provides a comparable window for observing recovery. These phase lengths were fixed prospectively rather than selected after inspecting the trajectories. They define a finite observation horizon and do not assume that stationarity is reached.

In each field-quench arm, the balanced vector of private fields changes sign at the first boundary. An agent previously receiving evidence favoring **amber**, for example, then receives evidence favoring **cobalt**, while the graph, model, and interaction rules remain unchanged. The agents are not told that a global intervention has occurred, so the collective response must develop from altered local evidence and neighbor-to-neighbor signaling. Matched nominal trajectories retain the initial field and control for changes that would occur without the quench. At the second boundary, the original fields are restored, allowing the experiment to measure whether collective organization returns, remains delayed, or persists over the available horizon. This is the logic of a finite-system quench and counter-quench, without assuming quantum dynamics or a thermodynamic limit.

Each model family contributes six independent graph/environment clusters. A cluster is a matched experimental replication block, not a community within the modular graph. It combines one graph realization with its private-field pattern, initial states, label mapping, and associated randomization. Each cluster contains four matched trajectories: nominal Markovized, field-quench Markovized, field-quench persistent memory, and field-quench scrambled history. All use $N = 16$, a reciprocal modular fixed-degree graph, coupling $J = 0.8$, and balanced disordered initialization. The modular topology provides local and community pathways through which signals can propagate, while matching graphs, fields, initial states, label maps, update orders, and other random variates reduces irrelevant between-condition variation. Because the four trajectories within a cluster share these design elements, the graph/environment cluster, rather than an individual agent or update, is the unit of statistical evidence.

3 Statistical-mechanical observables and inferential design

3.1 Macroscopic and information-theoretic observables

Belief and action order and their overlap are,

$$m_b = \frac{1}{N} \sum_i b_i, \quad m_a = \frac{1}{N} \sum_i a_i, \quad q_{ba} = \frac{1}{N} \sum_i b_i a_i. \quad (4)$$

Disagreement is the fraction of connected pairs with unequal states. For a symmetric comparison layer $W_s = (W + W^\top)/2$, we define

$$H_{\text{ref}} = -\frac{J_b}{2} \mathbf{b}^\top W_s \mathbf{b} - \frac{J_a}{2} \mathbf{a}^\top D_s \mathbf{a} - K \mathbf{a}^\top \mathbf{b} - \mathbf{h}^\top \mathbf{b} - \mathbf{g}^\top \mathbf{a}. \quad (5)$$

We report $e_{\text{ref}} = H_{\text{ref}}/N$. Equation (5) is an effective compatibility coordinate anchored in a symmetric reference model. We do not infer a Gibbs law for the LLM process and do not call this literal energy. Likewise $N \text{Var}(e_{\text{ref}})$ is an energy-fluctuation observable, not heat capacity unless an independently justified effective temperature exists. The belief susceptibility is reported without physical-temperature normalization,

$$\chi_b = N \text{Var}(m_b), \quad (6)$$

within a prescribed rolling window.

For discretized collective words x in a rolling window, Shannon configuration entropy is

$$S_{\text{config}} = - \sum_x \hat{p}(x) \log \hat{p}(x). \quad (7)$$

This is the usually used Shannon functional [29]. The entropy rate is a bounded-history plugged in estimate of the next word unpredictability. Single-variable marginal entropy measures local occupancy, pairwise mutual information measures dependence between variables. Total correlation [30],

$$\mathcal{T} = \sum_{k=1}^{2N} H(X_k) - H(X_1, \dots, X_{2N}), \quad (8)$$

separates marginal uncertainty from synchronous joint dependence.

The plug in joint entropy of 32 binary variables in a short window has a large finite-sample floor [31]. We therefore retain the raw estimate but independently circular-shift each variable 200 times. Shifts preserve each marginal occupancy and within-variable temporal ordering while breaking the synchronous cross variable relations. We report raw, null mean, their untruncated difference, and a normalized difference divided by the sum of marginal entropies. The same audit is applied to pairwise and edge mutual information. A negative adjusted estimate remains negative. The primary five-sweep representation is

$$Z_t = (m_b, m_a, q_{ba}, e_{\text{ref}}, v_E, S_{\text{config}}, h_{\text{rate}}, \mathcal{T}_{\text{adj}}, I_{\text{pair,adj}}, I_{\text{edge,adj}}, \chi_b, \rho_b, D_b)_t. \quad (9)$$

The three and seven sweep windows are prespecified sensitivities. This is an observable reduced representation, not a complete thermodynamic state.

3.2 Temporal asymmetry, spatial correlation, and persistence diagnostics

For a discrete observed word sequence and block length L , let $\hat{P}_L(x_0, \dots, x_L)$ be the regularized empirical forward block law. We compute

$$\mathcal{I}_L = \frac{1}{L} D_{\text{KL}} \left[\hat{P}_L(x_0, \dots, x_L) \parallel \hat{P}_L(x_L, \dots, x_0) \right] \quad (10)$$

in nats per attempted update. The primary $L = 3$ estimator uses pseudocount 0.5. Values at $L \in \{2, 4\}$ and pseudocounts 0.1 and 1.0 are sensitivities. Five hundred time permutations provide a panel-specific finite-length floor, and the adjusted statistic is the observed value in equation (10) minus its mean floor. Matched arms share the same shuffle tape. The machine-readable sensitivity table retains both the empirical permutation-null interval and the Monte Carlo standard error of its mean. These quantify different objects, dispersion of the null statistic and numerical uncertainty in the estimated bias floor, respectively. The added uncertainty columns reproduce, rather than replace, the prespecified floor mean and adjusted contrast.

For a fully observed stationary Markov chain with transition kernel P and stationary law π , the current

$$J_{xy} = \pi_x P_{xy} - \pi_y P_{yx} \quad (11)$$

can define an exact Markov entropy-production rate under the usual support conditions. We do not apply that interpretation to projected LLM trajectories. Persistent histories, semantic content, and bounded observations can violate first-order closure. Equation (10) is instead a coarse-grained time-reversal diagnostic and, under additional assumptions, may be a lower bound on hidden irreversibility [21, 23, 32].

Three secondary observables resolve spatial organization, temporal persistence, and the shape of the finite-size order-parameter distribution. First, for a phase window W we define the connected belief correlation at graph distance d ,

$$C_b(d; W) = \frac{1}{|\mathcal{P}_d|} \sum_{(i,j) \in \mathcal{P}_d} [\langle b_i b_j \rangle_W - \langle b_i \rangle_W \langle b_j \rangle_W], \quad \mathcal{P}_d = \{(i, j) : i < j, d_G(i, j) = d\}. \quad (12)$$

Shortest paths use the actual reciprocal modular delivery support. The bracket averages post-update observations at attempted-update resolution within one phase of one trajectory, node pairs are averaged inside that trajectory and never counted as independent replicates. Second, the normalized magnetization autocorrelation and its fixed truncation are

$$\begin{aligned} \rho_m(\ell) &= \frac{\langle [m_b(t) - \bar{m}_b] [m_b(t + \ell) - \bar{m}_b] \rangle}{\langle [m_b(t) - \bar{m}_b]^2 \rangle}, \\ \tau_{\text{int}}^{(\ell_{\text{max}})} &= \frac{1}{2} + \sum_{\ell=1}^{\ell_{\text{max}}} \rho_m(\ell). \end{aligned} \quad (13)$$

The primary descriptive truncation is $\ell_{\text{max}} = 2N = 32$ attempted updates, N and $3N$ are fixed sensitivities. Zero-variance phase series are left undefined rather than assigned artificial persistence. The quantity is a finite-window correlation sum reported in attempted-update units, not evidence of critical slowing down or an effective sample-size estimate. Its absolute shorter lag scale includes persistence induced by changing at most one agent on each random-sequential update; matched arms share that update convention. Quench and restoration windows can also be nonstationary, so their sums are finite-window persistence diagnostics rather than equilibrium integrated correlation times.

Third, we calculate the Binder cumulant

$$U_4 = 1 - \frac{\langle m_b^4 \rangle}{3 \langle m_b^2 \rangle^2} \quad (14)$$

separately for each phase and trajectory [33]. Near-zero second moments are reported as undefined. With only $N = 16$, equation (14) is a finite-size distribution-shape diagnostic; no crossing, critical point, or thermodynamic-limit transition is inferred. A post-reconstruction estimator sensitivity repeats the calculation over each full phase and its early and late halves. It compares the mean of the six trajectory-level cumulants with a pooled-moment construction; both uncertainty calculations resample complete graph-environment clusters, never updates.

3.3 Recovery estimands and exact inference

For each hold out graph environment cluster and model, the nominal manifold is fitted only from nominal trajectories in the other five clusters. Training medians impute nonfinite coordinates; training means and scales standardize features, shrinkage covariance with a prespecified ridge fraction defines a Mahalanobis like distance. The nominal 95th percentile

threshold is computed from those training distances. The hold out cluster contributes neither a scale nor a threshold.

We summarize peak departure, integrated response, path length, final residual, and threshold re-entry. The prospective cross model recovery contrast is

$$\Delta d_{\text{rec}} = \bar{d}_{31:35} - \bar{d}_{41:45}, \quad (15)$$

which can be positive, zero, or negative. It replaces a historical maximum minus final window quantity whose sign was structurally constrained. A complete graph/environment trajectory cluster is the inferential unit.

The hypotheses were specified before the cross-model outcomes were examined. H1 compares Granite field quench peak departure with matched nominal evolution at a one-sided $\alpha = 0.02$. H2 compares persistent with Markovized adjusted reversal divergence. H3 compares persistent with scrambled history. H4 tests the fixed recovery contrast. H2 to H4 form a one sided Holm family with total $\alpha = 0.03$. Exact cluster sign flips are the primary tests, deterministic 10,000 cluster bootstraps give intervals. The sign flip enumeration is exact conditional on the observed paired magnitudes under a sign-symmetry null; it is not described as treatment label randomization. Agent and time-point counts do not increase the inferential sample size. The exact test, multiplicity, and cluster bootstrap details are retained in Appendix A.

4 Study design and evidential roles

4.1 Exploratory discovery and corrected earlier evidence

The memory evidence developed in three stages, an exploratory experiment, a subsequent prospective replication, and the present cross-model extension with an additional history control.

The evidence for a memory effect was accumulated in distinct stages with different inferential roles. The exploratory stage identified increased temporal asymmetry under persistent history as a candidate phenomenon, while the prospective replication tested a contrast specified after that discovery. The later cross-model experiment then examined the effect under the frozen design and matched Markovized and scrambled-history controls for Granite and Qwen. Figure 2 presents these stages separately, with each point representing the effect observed within one independent graph/environment cluster and intervals constructed by resampling complete clusters. This separation allows the reader to assess consistency across stages and heterogeneity between model families without allowing the hypothesis-generating or earlier replication data to narrow the cross-model confirmatory interval retroactively.

The audit of the earlier quench experiment preserves every recorded prompt, completion, categorical choice, and trajectory. The original analysis computed a field panel’s recovery threshold from its own baseline, despite protocol text specifying a training-only threshold. The corrected analysis passes an explicit cluster-to-training-threshold map from the ‘leave one cluster out’ fitting into the recovery summaries.

The earlier study’s third hypothesis used the maximum recovery distance minus the final-five mean. That value remains numerically reproducible, but the construction is nonnegative for nearly every nonconstant trajectory. Its raw and Holm-adjusted sign-flip probabilities are retained as historical audit values and removed from inferential support.

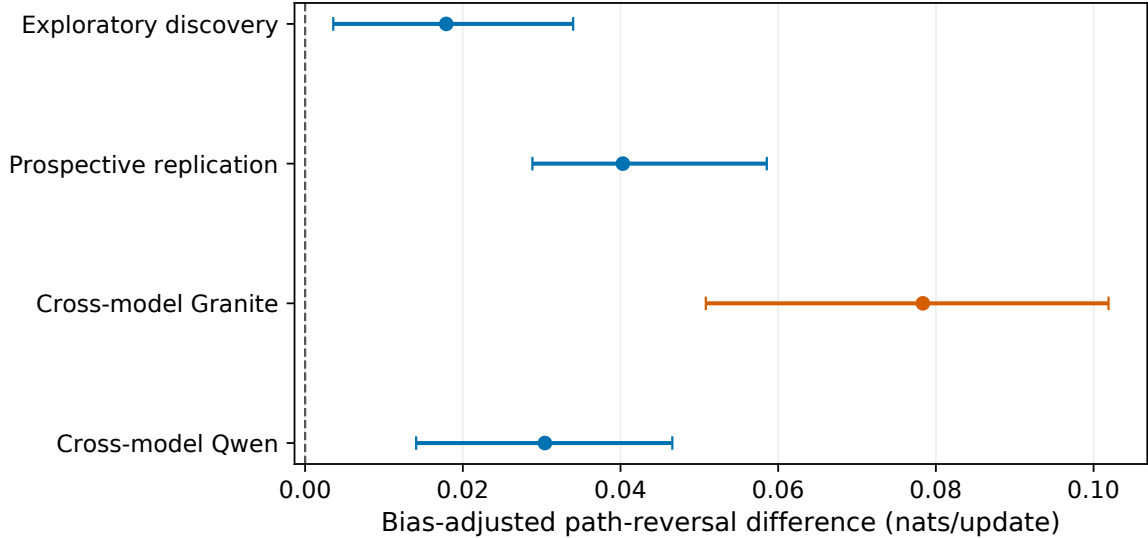


Figure 2: Exploratory discovery, prospective replication, and cross-model memory contrasts are shown separately. Points are graph/environment-cluster effects; intervals resample complete clusters. The exploratory and replication-stage estimates are not retroactively pooled into the cross-model confirmatory interval.

The corrected report distinguishes whether the numerical criterion was met, whether a directional test is valid, and whether the time-resolved path is consistent with return.

Figure 3 displays the corrected time series from the earlier quench experiment and places the nominal and field-reversal trajectories on a common time axis. This comparison distinguishes changes associated with the reversed private fields from background evolution that also occurs without an intervention. The yellow interval identifies the period during which the fields are reversed, while the dotted boundary marks their restoration and therefore separates the response phase from the recovery phase. The displayed observables provide complementary views of the dynamics: effective reference energy summarizes interaction compatibility, entropy measures configuration diversity, and distance quantifies displacement from nominal behavior in the cluster-excluded geometry. Together, the panels show whether the perturbation produces a coordinated departure during the quench and whether that departure contracts, persists, or changes form after restoration. The correction applies to the derived analysis used to construct these summaries; the underlying agent trajectories are unchanged. Cluster-level recovery and delayed-audit material is retained in Appendix A, including Figures 9 and 10.

4.2 Cross-model confirmation and control logic

To test whether the observed behavior extended beyond the model family in which it was first identified, the cross-model study used Qwen/Qwen2.5-7B-Instruct as the original family and IBM Granite-3.3-8B-Instruct as an independently developed comparison family [34, 35]. Both models were pinned to immutable revisions, with the full revision identifiers recorded in Appendix D and the repository manifests. Using two separately developed instruction-tuned families provides a stronger test of generality than repeating the experiment with another checkpoint from the same family. At the same time, the inference settings were held fixed: both models used the same pinned software stack, NF4 double quantization, BF16 computation, decoding temperature 0.5, top- $p = 0.9$, and a maximum of 96 generated tokens.

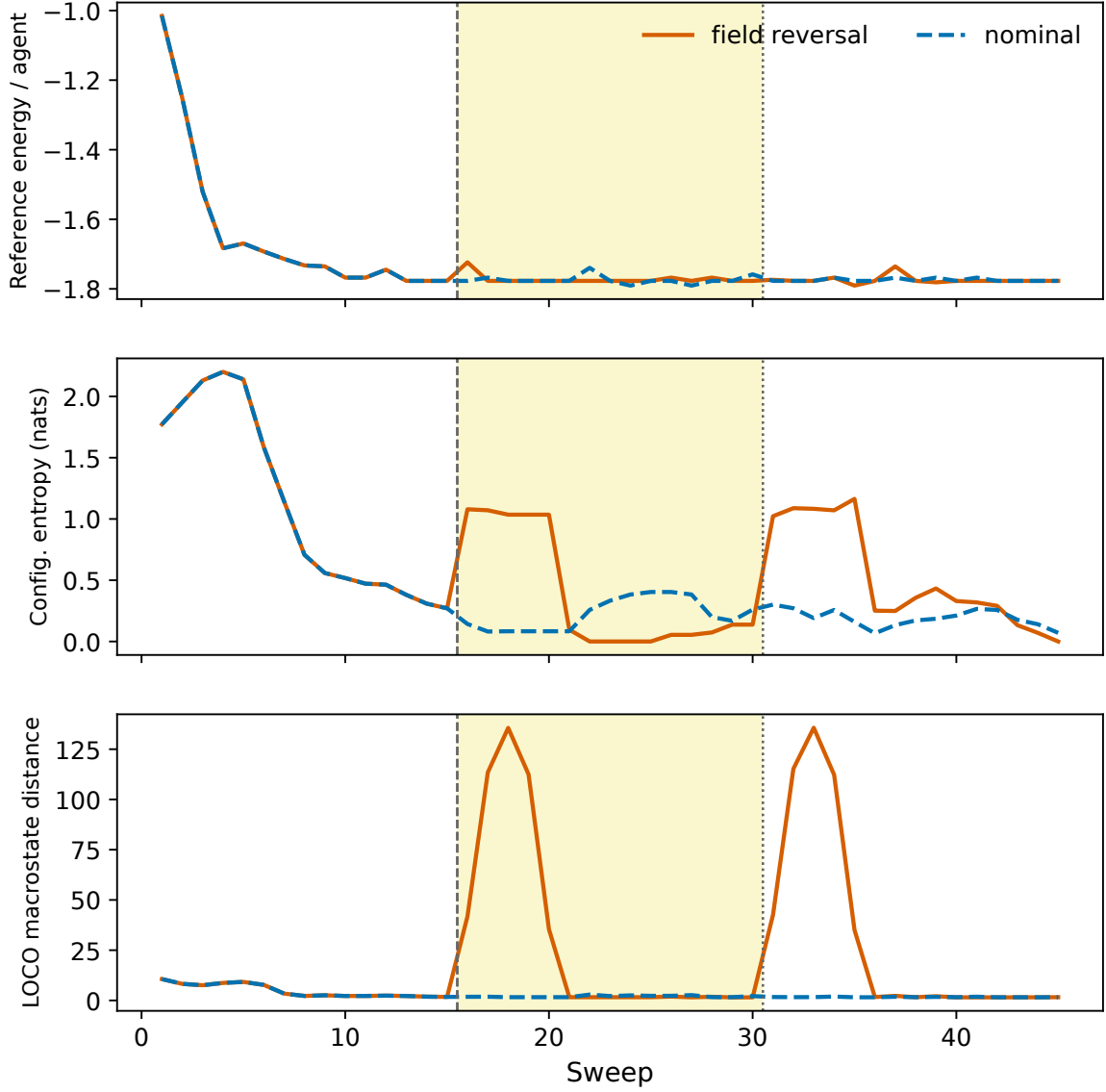


Figure 3: Corrected time series from the earlier experiment for nominal and field-reversal trajectories. The yellow interval marks the quench; the dotted boundary marks restoration. Reference energy is effective, entropy is configuration diversity, and distance is measured in the cluster-excluded nominal geometry.

The decoding temperature is a stochastic text-generation setting and is not interpreted as a physical or inferred thermodynamic temperature.

Before outcome evaluation, a limited engineering pilot verified that both families could execute the same protocol reliably. These checks covered model loading, JSON-schema validity, state occupancy, both transition directions, information isolation, packet delivery, prompt-token balance, latency, and projected compute requirements. The pilot did not calculate quench, memory, or block-reversal contrasts and therefore supplied no outcome evidence. The prospectively specified experiment then comprised 48 trajectories and 34,560 attempted local decisions across the two model families. Every complete trajectory was retained, including trajectories with weak, null, or unfavorable effects, so inclusion in the confirmatory analysis did not depend on the observed result.

5 Quench response, restoration, and finite-size organization

5.1 Cross-model response and recovery

The prespecified quench-response test (H1) was supported in the independent Granite family. The mean field-minus-nominal peak-distance contrast was 42.263 within-model distance units, with a 95% interval of [24.316, 59.303]. A positive contrast means that reversing the private fields drove the population farther from its nominal trajectory than would be expected from unperturbed variation. All six Granite cluster effects were positive. The separately retained Qwen results showed the same direction in all six clusters, with a mean field-minus-nominal peak contrast of 112.691 units. Complete cluster results were retained regardless of their direction or magnitude.

Figure 4 shows the nominal and field quench trajectories separately for the two model families. Within each panel, the figure shows when the quenched population departs from its matched nominal evolution and how that separation changes after the original fields are restored. The vertical scales should not be compared directly between Qwen and Granite because the nominal geometry, covariance, scaling, and thresholds are estimated separately within each model. These quantities are fitted while excluding the displayed cluster, so a cluster does not define the geometry in which its own response is measured.

The prespecified restoration test (H4) was also supported. The fixed restoration contrast was 52.541 distance units, with a 95% interval of [35.406, 68.673]. Its positive value indicates that distance from the nominal regime declined between the early and late restoration windows, and this decline occurred in every model to cluster pair. Unlike the historical recovery statistic used in the earlier experiment, this contrast was defined in advance and permits either sign.

Movement toward the nominal regime does not necessarily imply complete return within the finite observation horizon. Every Qwen trajectory re-entered its threshold, estimated from nominal training data, six sweeps after restoration. Only one of the six Granite trajectories re-entered by sweep 45; the remaining five had final five sweep means above their corresponding thresholds. Granite therefore showed clear movement toward restoration together with slower or incomplete finite-horizon recovery. The quench and counter quench paths need not retrace one another because messages, actions, and histories accumulated during the intervention alter the state from which restoration begins. These recovery measurements are finite time trajectory descriptors, not claims of equilibrium relaxation.

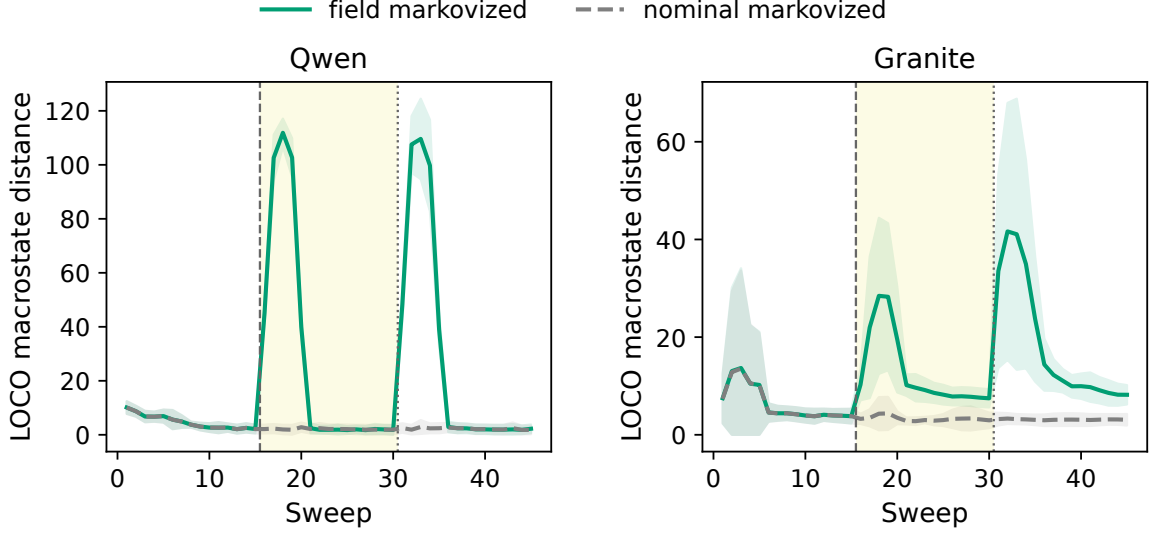


Figure 4: Model-specific field-quench and nominal trajectories. Curves are means over six graph/environment clusters and bands show between-cluster dispersion. Nominal fits, scaling, covariance, and thresholds exclude the displayed cluster.

5.2 Spatial and finite-window organization

The preceding response measures establish whether the population moves away from nominal behavior, but they do not show how that change is organized across the network or how long the resulting collective state persists. Connected graph-distance correlations address the first question by measuring whether neighboring agents reorganize together and whether that organization extends across the predefined communities. Finite-window autocorrelation and Binder statistics address the second by describing persistence and the shape of the observed order-parameter distribution. These are secondary descriptive analyses of the same recorded trajectories: they introduce no additional simulation arms and do not modify the confirmatory estimands. Because they were specified during reconstruction after the confirmatory aggregate results were known, they are not treated as a new prospective hypothesis family.

Connected correlation matrices from equation (12) are calculated separately within every complete model-graph-environment trajectory and phase. Their graph-distance profiles are then summarized over the six independent clusters. Figure 5 shows the cluster-averaged matrices for the persistent-history quench window together with phase-resolved correlation profiles by graph distance. The community boundaries are fixed by the graph construction rather than selected from the observed correlations, so the analysis asks how behavior aligns with a predefined interaction structure.

At graph distance one, the persistent-history quench-minus-baseline change in connected correlation was 0.00238, with a 95% interval of $[-0.01750, 0.02277]$, for Qwen and -0.02032 , with a 95% interval of $[-0.05486, 0.00947]$, for Granite. Both intervals include zero, indicating that the available clusters do not resolve a consistent nearest-neighbor change during the quench. These intervals describe finite-window spatial reorganization and are not assigned a confirmatory disposition.

The finite-window diagnostics examine temporal organization from a complementary perspective. Appendix C and figure 14 combine the full autocorrelation curve with the fixed-lag integral in equation (13), the Binder statistic in equation (14), and empirical

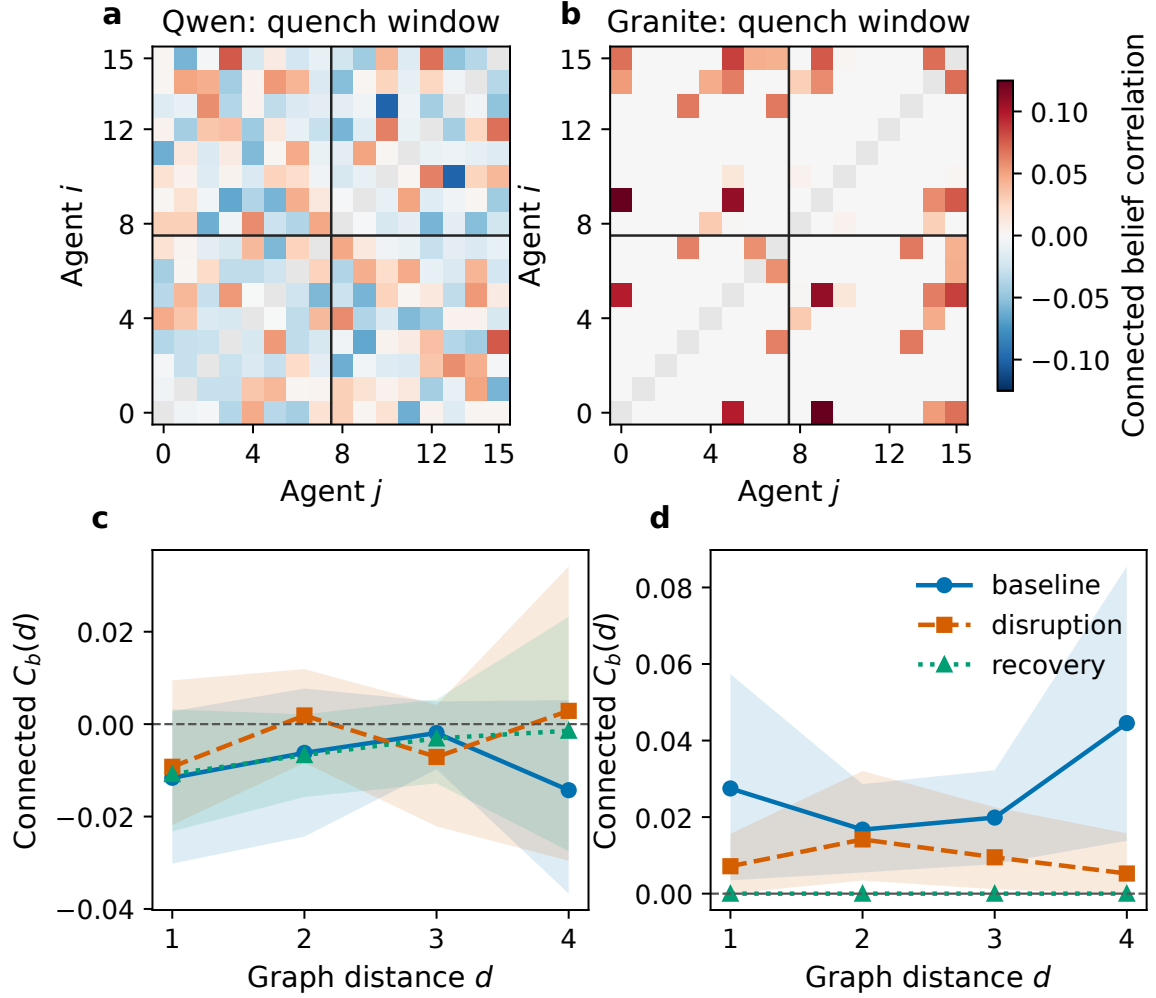


Figure 5: Connected belief correlations in the persistent-history arm. The upper panels average complete 16×16 connected-correlation matrices over all six clusters in each model; black rules mark the two predefined communities. Agent indices align across graph realizations only through those community memberships, so fine node-level texture in an averaged matrix has no cross-realization identity interpretation. The lower panels show equation (12) by graph distance, with 95% complete-cluster bootstrap intervals. Individual pairs and updates do not enter the inferential sample size.

magnetization occupancy. Showing occupancy alongside the Binder statistic is important because a finite-window Binder value can be strongly affected by the limited distribution of observed magnetization states. Autocorrelation also measures a different property from block-reversal divergence: autocorrelation is a time-symmetric measure of two-point persistence, whereas block reversal compares the frequencies of forward and reversed path words.

At the fixed two-sweep truncation, the persistent-minus-Markovized recovery change in τ_{int} was -2.46995 updates, with a 95% interval of $[-5.79214, 0.17724]$, for Qwen and -4.14109 updates, with a 95% interval of $[-19.88286, 11.60067]$, for Granite. The field-Markovized quench-minus-baseline Binder change was -4.80504 , with a 95% interval of $[-7.42824, -2.20832]$, for Qwen and 0.16643 , with a 95% interval of $[-0.09447, 0.52063]$, for Granite. The accompanying source table retains full-window, early-half, and late-half Binder estimates under both cluster-first and pooled-moment constructions. Differences among these estimates are used to diagnose finite-window and aggregation sensitivity, not to select the construction that produces the largest separation. All results are reported regardless of sign or compatibility with zero, and none is interpreted as evidence of an equilibrium correlation time, critical slowing down, a Binder crossing, or a phase transition.

6 Memory and projected temporal asymmetry

6.1 Persistent, Markovized, and scrambled histories

The three history conditions distinguish between having no explicit personal history, having genuine history, and merely having additional history-shaped text. In the Markovized condition, an agent sees its current state but not its earlier decisions. In the persistent condition, it also sees its last three genuine belief, action, and memory records. The scrambled condition provides a similarly formatted history section, but its contents are randomized and no longer correspond to the agent’s actual past.

Block-reversal divergence measures whether short sequences of projected population states occur differently from the same sequences read in reverse. A larger value indicates a stronger temporal direction in this reduced description of the trajectory. Relative to the Markovized condition, genuine history increased the bias-adjusted block-reversal divergence by 0.05438 nats per attempted update, with a 95% interval of $[0.03459, 0.07548]$. This prespecified comparison, H2, was supported. It shows that providing an agent with its actual recent history changes the temporal organization of the observed population dynamics beyond what is produced by a current-state prompt alone.

The comparison with scrambled history provides a stricter control. Genuine history increased adjusted block-reversal divergence by 0.04845 nats per attempted update relative to the scrambled condition, with a 95% interval of $[0.02190, 0.07526]$. This prespecified comparison, H3, was also supported. Because both prompts contain similarly formatted history sections, the result indicates that the relationship between the displayed history and the agent’s actual past matters beyond the mere presence or approximate length of additional prompt text.

Some adjusted values for individual arms are negative because the correction subtracts the average divergence expected from the finite-sample shuffle baseline. An observed value below that baseline therefore produces a negative adjusted score. This does not represent negative physical entropy production, and the values are not truncated to zero before computing the paired H2 and H3 contrasts. The positive contrasts show a difference

between history conditions, not that either arm has positive absolute thermodynamic entropy production.

Figure 6 displays the paired cluster-level contrasts for the two model families. Each point represents one independent graph/environment-cluster comparison, so its position relative to zero shows whether genuine history increased or decreased adjusted block-reversal divergence within that matched cluster. The separate model summaries also make visible whether the size and consistency of the effect differ between Qwen and Granite.

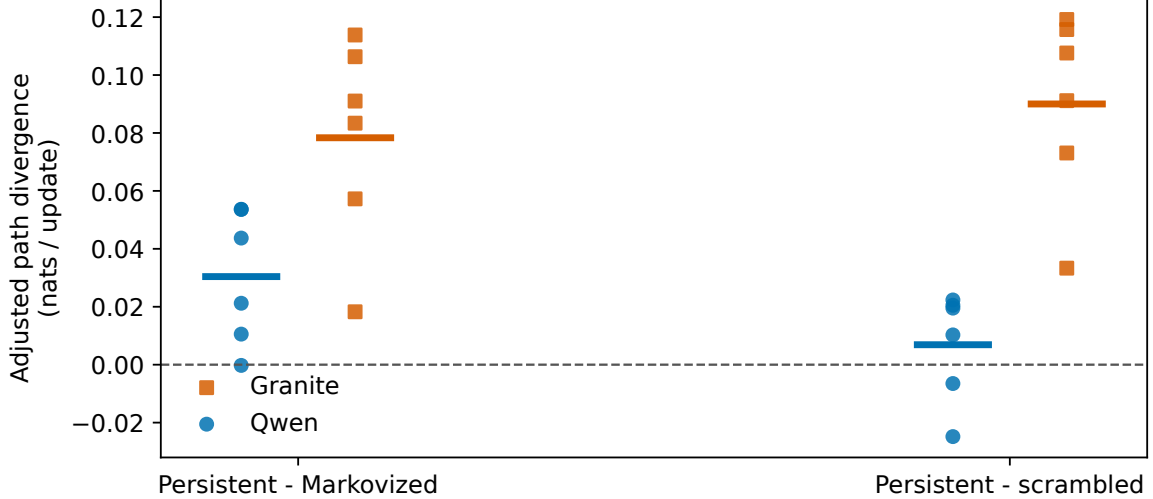


Figure 6: Per-cluster genuine-memory contrasts for both pinned model families. Short horizontal segments indicate model means. The scrambled-history arm controls prompt section and approximate length while destroying temporal relation to the current trajectory.

No exact thermodynamic interpretation follows from these results. The block words summarize selected categorical coordinates and omit the natural-language messages and part of each agent’s private state. Sensitivity tables for block length, pseudocount, conditional memory depth, state occupancy, and prompt balance test how strongly the result depends on these analytical choices. The prompt-balance diagnostic in figure 12 of Appendix B records the scope of the format and approximate-length control.

6.2 Model heterogeneity and estimator dependence

The pooled H2 and H3 results show that genuine history changes the projected dynamics on average across the prespecified model stratified clusters. A supported pooled effect, however, does not imply that Qwen and Granite respond to history in the same way or with the same magnitude. It also does not imply that every reasonable construction of the block reversal estimator produces an identical result. We therefore examine model heterogeneity and estimator sensitivity separately to define the scope of the supported memory findings.

For persistent relative to Markovized prompting, the model-specific mean contrast was 0.03041 nats per attempted update for Qwen, with five of six cluster effects positive, and 0.07835 for Granite, with all six positive. Both families therefore showed the same average direction for H2, although the effect was larger and more consistent in Granite. For persistent relative to scrambled history, the mean contrast was 0.00689 for Qwen, with four of six cluster effects positive, and 0.09001 for Granite, with all six positive. The separate Qwen scrambled-control sign-flip value was 0.21875 and was not itself a confirmatory test. The prespecified pooled H3 remained supported because its inferential units were the ‘model

stratified’ cluster pairs. The appropriate conclusion is therefore that temporally related history has a supported pooled effect, with a strong Granite contribution, rather than that the content-specific effect is uniform across model families.

The magnitude of the memory contrast also depends on how block-reversal divergence is estimated. Block length determines how much temporal structure is represented, the pseudocount controls sparse word frequencies, and the shuffle floor corrects finite-sample bias. The pooled memory contrasts remain positive across most prespecified combinations of these choices, but their magnitudes vary substantially. At block length four and pseudocount one, for example, the persistent-minus-scrambled contrast falls to 0.00024 nats per attempted update, and the Granite component becomes slightly negative. Conditional-memory information is also not consistently larger in the persistent condition under every construction.

These sensitivity results do not reverse the prespecified H2 or H3 dispositions. They limit the claim to the prespecified coarse graining, bias-floor correction, and paired controls, rather than supporting an estimator-independent entropy-production rate. The block-length, pseudocount, and shuffled-floor results are displayed in Figure 11 of Appendix B.

Taken together, H1 to H4 addresses four distinct questions. H1 tests whether the field reversal drives the system beyond nominal evolution, H2 tests whether genuine history matters relative to current-state prompting, H3 tests whether temporally related history matters beyond a format-matched scrambled control; and H4 tests movement toward the nominal regime during the fixed restoration window. Their joint support is therefore a synthesis of complementary results, not evidence for a single universal effect.

Figure 7 brings the four prespecified tests together as a summary of the main confirmatory findings. These tests address distinct questions rather than repeated measurements of one underlying effect. H1 asks whether the field quench produces a departure beyond nominal evolution, while H4 measures movement back toward the nominal regime between the early and late restoration windows. H2 compares genuine persistent history with current-state-only Markovized prompting, and H3 compares genuine history with the format-matched scrambled-history control. The left panel displays H1 and H4 in within-model distance units, whereas the right panel displays H2 and H3 in nats per attempted update. In each case, the point is the estimated effect, the horizontal bar is its 95% complete-cluster bootstrap interval, and the dashed vertical line marks zero. All four intervals lie on the positive side of zero, but positivity has a different meaning for each test: greater quench-induced departure for H1, greater early-to-late restoration for H4, and greater adjusted block-reversal divergence under genuine history for H2 and H3. The separate panels prevent distance and information quantities from being compared as though they shared a common scale or effect magnitude. Under the prespecified tests, exact cluster sign-flip calculations, and multiplicity allocation, all four criteria were met. The figure therefore provides a compact synthesis of the supported outcomes, but it does not combine them into a single composite effect. The model-specific differences, finite-window interpretation, and estimator limitations described above remain part of the result.

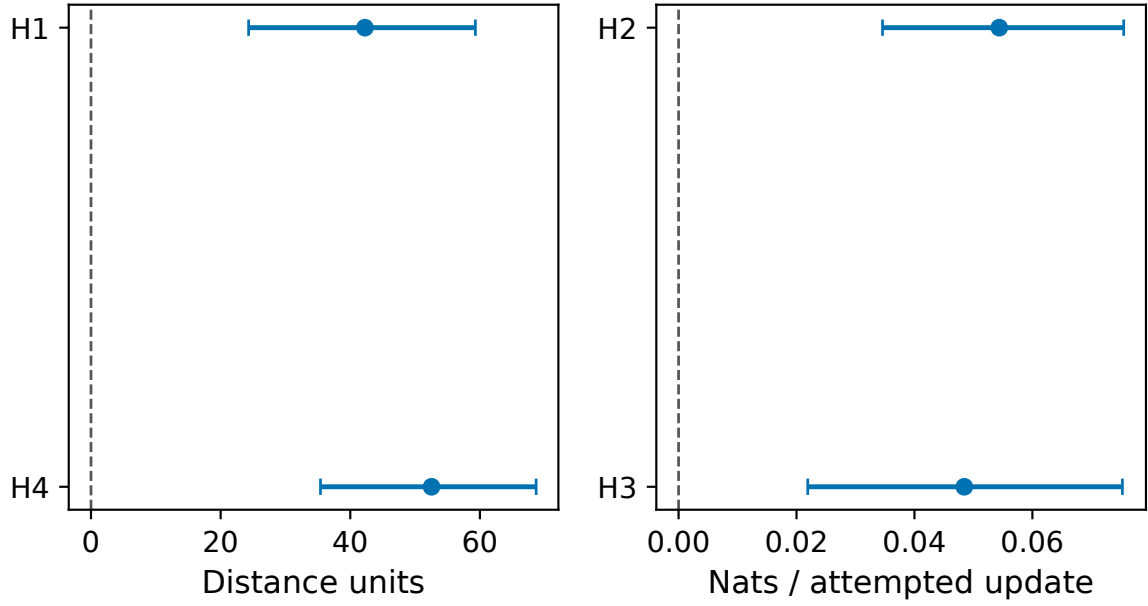


Figure 7: Four prespecified effects with 95% complete-cluster bootstrap intervals. The left panel shows the field-quench response (H1) and fixed-window restoration contrast (H4) in within-model distance units. The right panel shows genuine persistent history relative to Markovized prompting (H2) and scrambled history (H3) in nats per attempted update. Points are estimated effects, horizontal bars are bootstrap intervals, and dashed vertical lines mark zero. Distance and information quantities are kept on separate axes because their numerical magnitudes are not comparable. Prespecified dispositions use exact cluster sign flips and the multiplicity allocation.

7 Reduced dynamical descriptions and closure limits

7.1 Information retained beyond mean order and the kinetic surrogate

Magnetization records directional order but can remain near zero when half the population reorganizes in opposing communities. Configuration entropy then records collective state diversity; marginal entropy records local occupancy; total correlation distinguishes independent uncertainty from coordinated dependence; effective energy records compatibility with the symmetric reference; susceptibility and energy variance record fluctuations; autocorrelation records finite-window persistence; and path reversal records temporal ordering. These quantities are complementary only when their estimators and scales are kept distinct.

The representation analysis of the earlier quench experiment uses transparent multinomial logistic regression with leave-one-cluster-out preprocessing. The delayed permutation audit tests both absolute balanced accuracy and the increment over order-only features. This is not a claim that entropy outperforms all ordinary summaries. It tests whether a predefined collective representation retains condition information discarded by a reduced representation. The field quench trajectory supplies a mechanistic example, energy, dependence, entropy rate, and fluctuation can change while mean magnetization remains balanced.

We fit logistic belief and action responses only to the immutable microscopic response table from the prospective replication stage. The belief model includes private field, delivered neighbor field times controlled coupling, previous belief, and previous action; the action model includes updated belief and previous action. No coefficient is refitted to either later quench trajectory set. The surrogate is run on the same graph generator, initial state generator, asynchronous update tape, field schedule, and intervention times. It is a deliberately low-dimensional kinetic Ising comparison rather than a fitted description of every asymmetric or history-dependent response [36].

Figure 8 compares the direct Qwen-agent trajectories from the earlier field-quench experiment with trajectories generated by the low-dimensional kinetic surrogate. The solid blue curves represent the observed LLM-agent dynamics, whereas the dashed orange curves show what the surrogate predicts when it is run with the same graph, initial conditions, update schedule, and field changes. This comparison is genuinely out-of-sample: none of the surrogate coefficients are fitted or adjusted using the quench trajectories displayed in the figure. The eight panels examine different aspects of the collective response, including belief and action magnetization, belief-action overlap, effective reference energy, configuration entropy, susceptibility, finite-window correlation time, and shared-response distance. The yellow region marks the field-reversal interval, with its boundaries identifying the quench and restoration events. A successful reduced description would not need to reproduce every microscopic fluctuation, but it should capture the direction, timing, relative amplitude, and recovery of the main collective changes across these observables. Agreement therefore identifies behavior that can be explained by the surrogate’s small set of local variables, while disagreement reveals collective, semantic, asymmetric, or history-dependent information that the reduced closure omits.

In the five coordinate comparison geometry, the direct field minus nominal peak response difference is only 0.142 units on average and is nonzero in one of six clusters, whereas the surrogate difference is 1.467 units and is positive in four clusters. The corresponding field-minus-nominal energy entropy route-area differences are 0.043 and 0.695. The direct shared response peaks at the quench boundary in one cluster and is otherwise dominated by

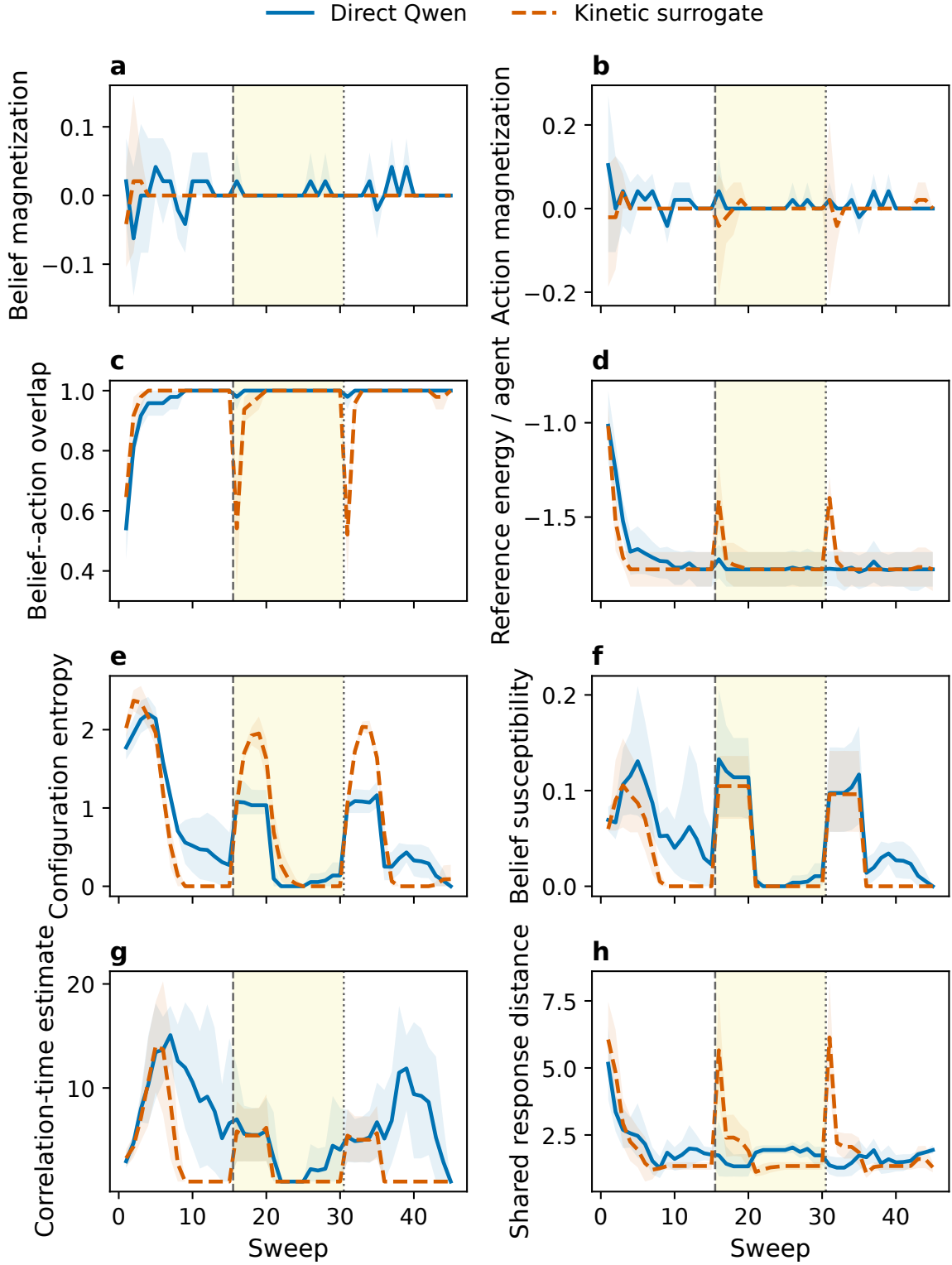


Figure 8: The out of sample field-quench comparison between direct Qwen trajectories and the replication stage kinetic surrogate. Matching direction, peak timing, normalized amplitude, recovery, and route area are assessed separately. Surrogate failure is evidence about the limits of the closure, not a failed LLM experiment.

initial relaxation; the surrogate places its peak at the quench or counter-quench boundary in four clusters. Both descriptions return close to their own final shared-response baseline, but the surrogate neither reproduces the direct amplitude nor its timing. This also explains why the richer macrostate distance used in the earlier quench analysis can show a large quench pulse that a short kinetic closure over a small shared coordinate set misses.

7.2 Captured response, failure modes, and finite-size context

This comparison identifies three possible failure modes. First, a homogeneous logistic neighbor coefficient cannot capture state-dependent semantic use. Second, action and belief persistence can introduce lags not represented by a single synchronous mean field. Third, hidden memory and workload can produce different future laws at the same projected spin configuration. Adding unconstrained parameters after observing quench trajectories would obscure these limits, so the closure remains low-dimensional and out of sample.

CPU trajectories at $N \in \{8, 16, 32, 64\}$ supply a dense effective-model size context. They are not direct LLM finite-size scaling and cannot establish a critical exponent or thermodynamic-limit phase transition. The direct $N = 16$ paths remain empirical anchors.

The effective-model size context is displayed in figure 13 of Appendix B.

8 Discussion

Treating the LLM-agent network as a stochastic interacting system begins with implemented rather than metaphorical objects, agent instances receive local prompts, exchange packets, update on a recorded schedule, and change categorical state. Magnetization, Shannon entropy, mutual information, total correlation, and path divergence are then mathematical observables of those records. State separation is central to this interpretation because it distinguishes the complete augmented state that determines a transition from the local information available to an agent and from the lower-dimensional projection available for population-level measurement.

The quench experiments use this formulation to measure finite-horizon response and recovery rather than equilibrium relaxation. Both model families depart from their within-model nominal geometries after field reversal, and the fixed-window restoration contrast supports motion back toward the restored nominal regime. The difference between complete Qwen threshold re-entry and slower or incomplete Granite re-entry within the observation horizon remains important: the common directional response does not imply a common recovery time or complete return. Recovery times and threshold crossings are trajectory descriptors, and the quench and counter-quench paths need not retrace because messages, actions, and history change the state reached at restoration.

The history interventions clarify what memory means under coarse-grained observation. Persistent history differs from an explicit current-state prompt, and comparison with scrambled history asks whether temporally related content matters beyond prompt format and approximate length. The distinction between complete augmented state, observable projection, and rolling macrostate explains how private memory can generate history dependence in the projected process. Positive paired contrasts nevertheless do not establish positive absolute entropy production: some arm level bias-adjusted values are negative, the observed path words omit natural language content and private state, and the measured reversal divergence is a coarse-grained projected statistic rather than an exact physical entropy production rate.

Model heterogeneity and negative findings limit generalization. The persistent versus Markovized contrast has the same direction across the two families, whereas the content specific scrambled history contrast is much weaker and less uniform for Qwen than for Granite. Estimator choices also change the magnitude of the memory contrasts. In the exploratory evidence, degree and traffic matched nonreciprocity did not produce a reliable increase in irreversibility, and partition and message corruption trajectories in the earlier quench experiment remained close to nominal under their tested construction. These results prevent the broader assertion that any directed graph, entropy statistic, or history prompt produces the same collective effect.

The secondary spatial, persistence, Binder, and closure analyses add distinct but limited information. Connected correlations describe how finite-window organization changes across the fixed modular graph, while autocorrelation summarizes persistence of the projected collective coordinate. Binder values are shown with occupancy and aggregation sensitivities, they do not establish a crossing, critical slowing down, or a phase transition. The out of sample kinetic surrogate captures some direction and recovery features but misses parts of the direct response amplitude and timing. Its inexpensive size context is not direct LLM finite size scaling, so surrogate failure identifies limits of the reduced closure rather than a failed agent experiment.

These measurements support an effective statistical mechanical description, not a literal identification of the network with a thermodynamic material. The reference energy is effective because the LLM transition rule is not derived from its Hamiltonian, and decoding temperature is a generation setting rather than an independently inferred thermodynamic temperature. No free energy like coordinate is emphasized because such a temperature is unavailable. The value of the framework is instead that it keeps collective organization, fluctuations, dependence, response, and temporal ordering operationally distinct while exposing the assumptions behind each reduction.

Several limitations bound the scope. Two 7B instruction model families do not establish that it will apply identically to every model, and the direct systems contain only $N = 16$ agents, so no thermodynamic limit transition is claimed. The binary projection compresses semantic messages and private state. The confirmatory study uses one reciprocal modular topology and one coupling/noise anchor. The scrambled history placebo matches format and approximate token distribution but cannot make semantic content exchangeable turn by turn. Short window entropy, dependence, autocorrelation, and path estimates remain sensitive to coarse graining and finite sampling even after their null floor and aggregation checks.

Further work therefore requires larger direct systems, additional network topologies, more model families, and longer stationary trajectories. Richer observable state representations could test which hidden variables account for projected memory, while prospectively chosen alternative coarse grainings could establish whether the present estimator dependence persists.

9 Conclusions

Interacting LLM agents provide a new class of finite many body system in which microscopic updates are language conditioned decisions rather than fixed spin-flip rules. Each agent acts from local evidence, delivered neighbor signals, and bounded private state, while collective organization emerges from repeated interaction across a network. This preserves the central statistical mechanical question of how microscopic interactions generate macroscopic

behavior, but places it in a system with contextual, history-dependent, and partially observed transition rules. The resulting dynamics can nevertheless be characterized through order, fluctuations, correlations, response, persistence, and temporal asymmetry. The principal statistical-mechanical contribution is therefore a reproducible micro to macro framework for formulating and measuring collective dynamics when the local transition law is generated by an LLM.

The controlled field quench protocol showed that these measurements reveal behavior that is not captured by final consensus or population means alone. Reversing the agents’ private evidence produced a supported collective response across two independently developed model families, while restoring that evidence revealed model dependent recovery over a fixed observation horizon. Persistent history also changed the temporal organization of the projected dynamics relative to both current state only prompting and a format matched scrambled history control. The latter comparison shows that the relationship between an agent’s recent history and its current decision matters beyond the generic effect of adding more structured text to a prompt. Spatial correlations, finite window diagnostics, and the out of sample kinetic surrogate further identified which aspects of the response admit a reduced description and which remain dependent on model family, history, or richer collective coordinates. The weaker Qwen scrambled history decomposition, Granite’s incomplete finite horizon recovery, and the estimator sensitivities define the boundaries of these supported findings rather than being hidden by the pooled results.

For agentic AI, these results show why multi-agent systems should be evaluated as dynamical populations rather than only through individual answer quality, task completion, or eventual agreement. Systems with similar average outcomes can differ in internal coordination, sensitivity to changing evidence, dependence on memory, propagation of local signals, and recovery after perturbation. Statistical mechanical observables therefore provide a useful basis for comparing agent architectures, communication rules, memory policies, and control interventions, while reduced models test whether those collective dynamics can be predicted from a manageable set of variables. Larger networks, additional topologies and model families, longer trajectories, and richer out of sample closures are natural next steps toward determining which features generalize. The interpretation remains effective rather than literal: decoding temperature is not physical temperature, reference energy is not microscopic energy, and projected block-reversal divergence is not exact thermodynamic entropy production. Within those boundaries, the framework provides a disciplined way to measure, compare, and ultimately control the collective behavior of interacting agentic AI systems.

Data and code availability

Code, simulation configurations, figure source data, aggregate results, and reproduction scripts are publicly available at <https://github.com/mantzaris/ThermoAgent>. The reported cross-model experiment was run on an NVIDIA GeForce RTX 4090 and required 19.19 measured generation GPU-hours.

Raw prompts, model completions, and complete trajectory records are not included in the public repository because of their storage volume and the potential sensitivity of generated content. The original external raw artifact tree is no longer retained. Content-addressed manifests, replay records, and aggregate-validation artifacts are provided to document the reconstruction and verify the reported aggregate results.

Acknowledgements

OpenAI Codex (GPT-5) was used for code assistance, the author reviewed all resulting material and takes responsibility for the manuscript.

A Study chronology, correction, and inferential details

The earlier quench analysis originally derived each field panel’s recovery threshold from that panel’s own baseline, although the protocol called for a threshold fitted only to the training clusters. The corrected analysis maps the leave-one-cluster-out training threshold into the held-out recovery summary without changing any recorded trajectory. Figure 9 retains every cluster-level recovery path, including the high-response cluster, under the corrected training-only construction. The historical maximum-minus-final recovery statistic remains numerically reproducible but is not used as directional evidence because its construction is nonnegative for nearly every nonconstant trajectory.

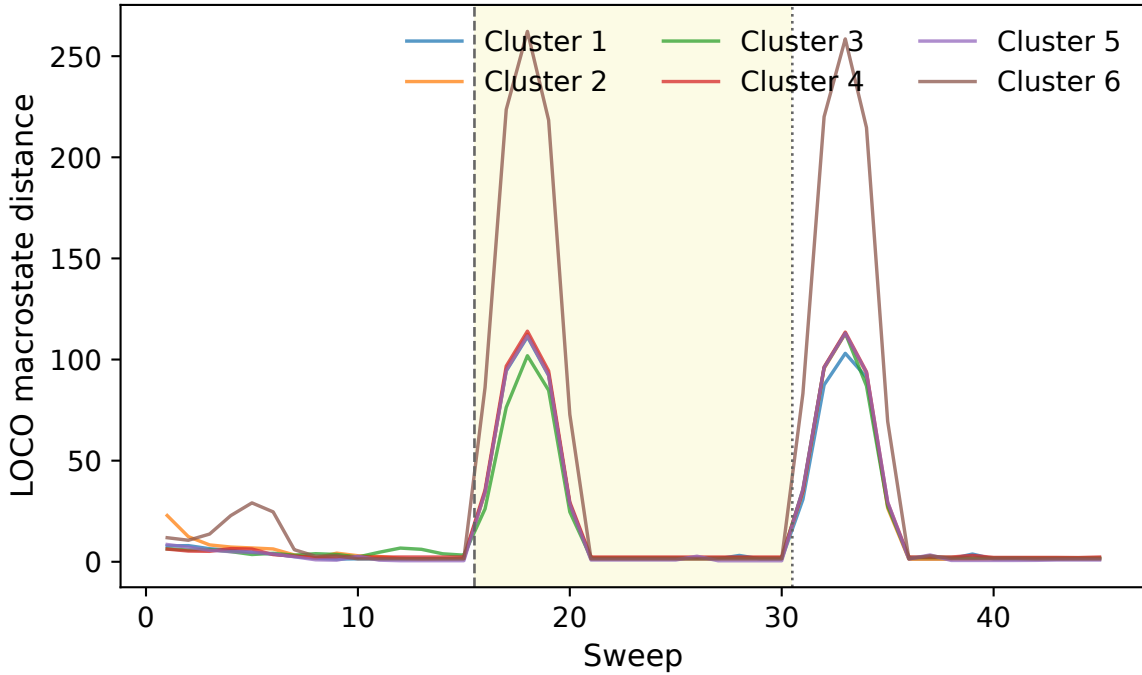


Figure 9: All field-reversal cluster trajectories from the earlier experiment, including the high-response cluster. The corrected audit replaces the held-out-panel threshold with the training-cluster nominal threshold. The historical recovery sign statistic is not used as directional evidence.

The delayed audit executes the omitted 10,000 cluster-preserving label permutations. All four panels stay together within each cluster, and every replicate refits imputation, scaling, logistic regression, and leave-one-cluster-out predictions. Three-, five-, and seven-sweep nominal geometries are fitted independently. Each coordinate and observable family is deleted in turn, and dependence estimates receive circular-shift null floors. These were delayed prespecified sensitivity analyses completed after the primary outcomes because they had been omitted from the initial implementation, not a new prospective experiment.

The representation permutation retains the four condition panels in every cluster and permutes labels only within cluster. For each of 10,000 replicates, every leave-one-cluster-out fold refits imputation, scaling, and the multinomial logistic model; test-cluster features do not enter preprocessing. Figure 10 records the delayed dependence and full-pipeline permutation audit.

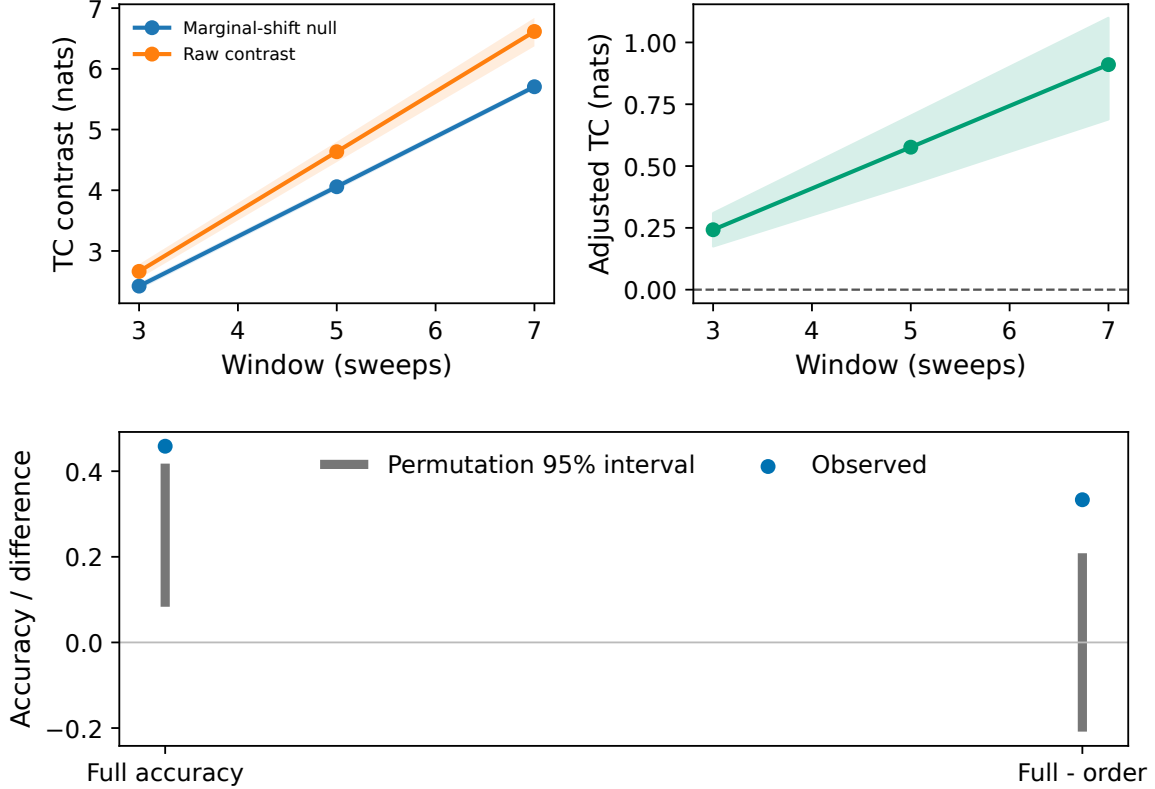


Figure 10: Delayed prespecified dependence and permutation audits. Raw and bias-adjusted dependence, window sensitivity, and cluster-preserving nulls are shown explicitly. These audits were completed after the primary outcomes because of implementation omissions.

For the prospective cross-model tests, the one-sided sign-flip probability enumerates all 2^n sign assignments of the observed cluster effects for $n \leq 20$ paired clusters. The cluster bootstrap samples complete graph/environment clusters with replacement and never resamples time points as independent observations. H1 receives alpha 0.02; H2–H4 receive a Holm-adjusted family alpha 0.03. The allocation was fixed before outcomes were examined. Intervals are uncertainty summaries and do not replace the exact paired tests.

B Estimator, control, and closure sensitivities

Total correlation nulls independently circular shift each coordinate by a nonzero offset where possible. This retains marginal occupancy and each coordinate’s internal temporal ordering while breaking synchronous dependence. The adjusted estimate is not truncated. Alternative rolling windows rebuild the words, covariance, training threshold, and held out distance rather than subsampling a five sweep output.

For block reversal, time shuffled floors quantify finite length occupancy bias, they do not prove stationarity. Results at block lengths two, three, and four expose finite history sensitivity, while conditional mutual information at history depths one through three

provides a separate diagnostic of incomplete first order closure. Figure 11 displays the prespecified estimator-sensitivity grid.

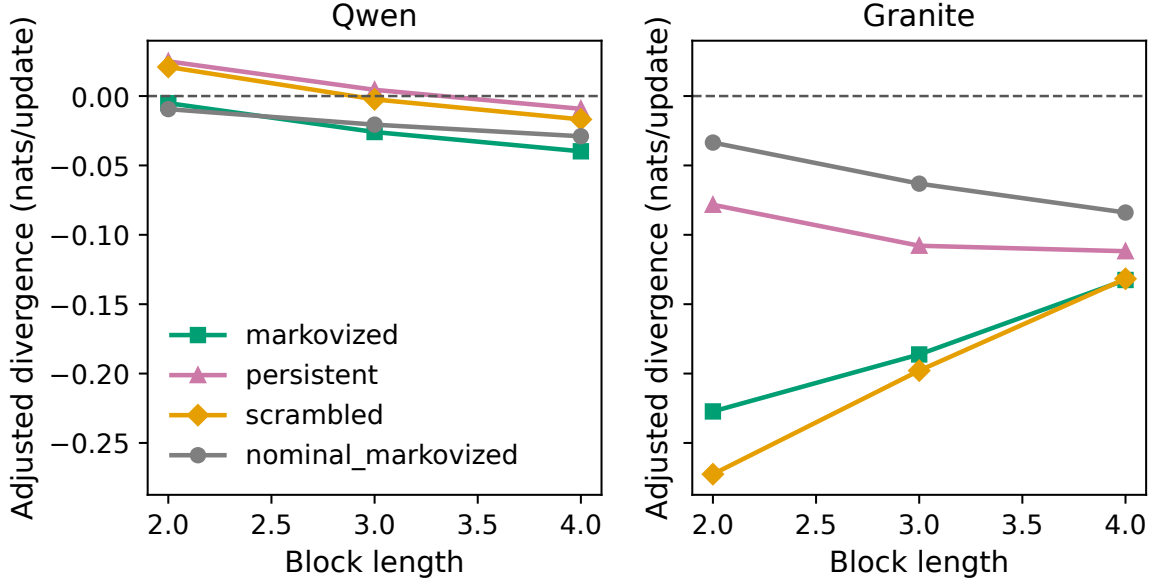


Figure 11: Block-length, pseudocount, and shuffled-bias-floor sensitivity for bias-adjusted block reversal divergence. Memory contrasts can be checked against estimator choices; observable coarse-graining and finite length remain limiting.

The scrambled-history arm controls the presence and approximate length of the history prompt while destroying its temporal relation to the current trajectory. Figure 12 records the token-count diagnostic for that comparison. It supports approximate prompt-length and format balance but does not establish turn-by-turn token identity or semantic exchangeability.

Finally, CPU trajectories at several sizes provide context for the reduced kinetic model. Figure 13 places the direct-system anchor beside that effective-model size comparison. These calculations are not direct-LLM finite-size scaling and do not support a thermodynamic-limit claim.

C Finite-window persistence and Binder diagnostics

The autocorrelation and Binder diagnostics defined in equations (13) and (14) are summarized in Figure 14. Each quantity is computed within one complete mode graph environment trajectory before uncertainty is summarized across the six clusters per model. The autocorrelation sum uses the fixed two sweep truncation, while one and three sweep truncations remain machine-readable sensitivity analyses. Zero-variance phase series are left undefined rather than assigned artificial persistence, and uncertainty resamples complete clusters rather than treating time points as independent.

The Binder statistic is likewise calculated trajectory first and describes the shape of a finite $N = 16$ magnetization distribution. The source data table retains full phase, early half, and late half estimates under both cluster mean and pooled moment constructions, near-zero second-moment cases remain undefined. Empirical magnetization occupancy is shown alongside the Binder values so that aggregation sensitive cumulants remain connected to the finite-window distributions from which they are calculated.

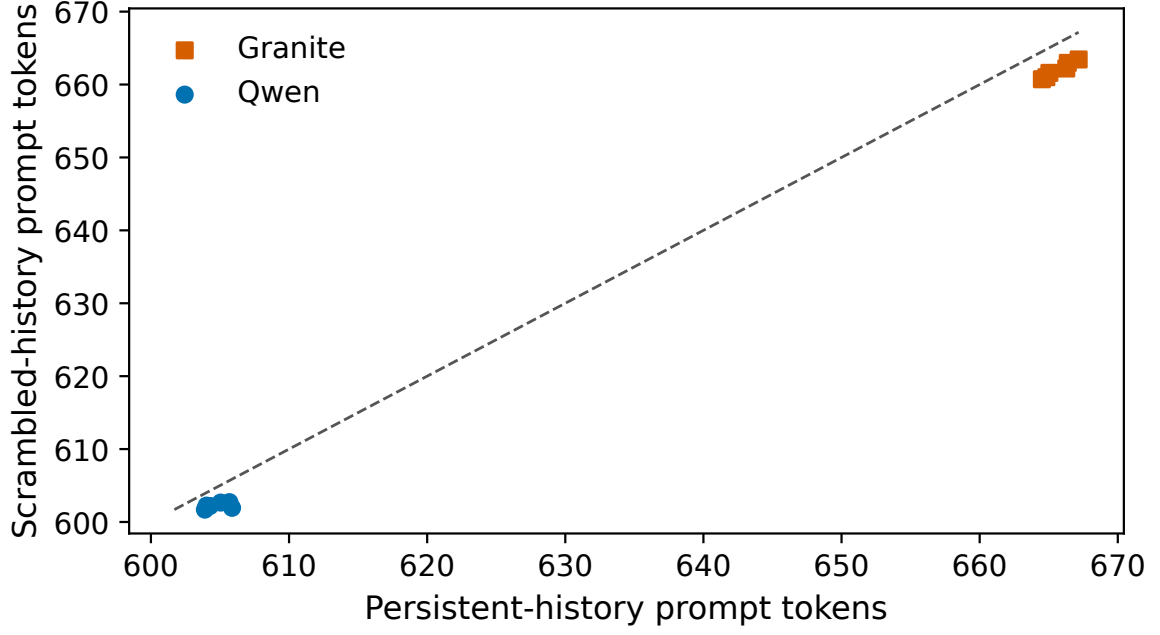


Figure 12: Scrambled-history prompt-length control. Per-cluster mean token counts verify that prompt length and format are approximately matched; semantic content cannot be exactly token-matched turn by turn.

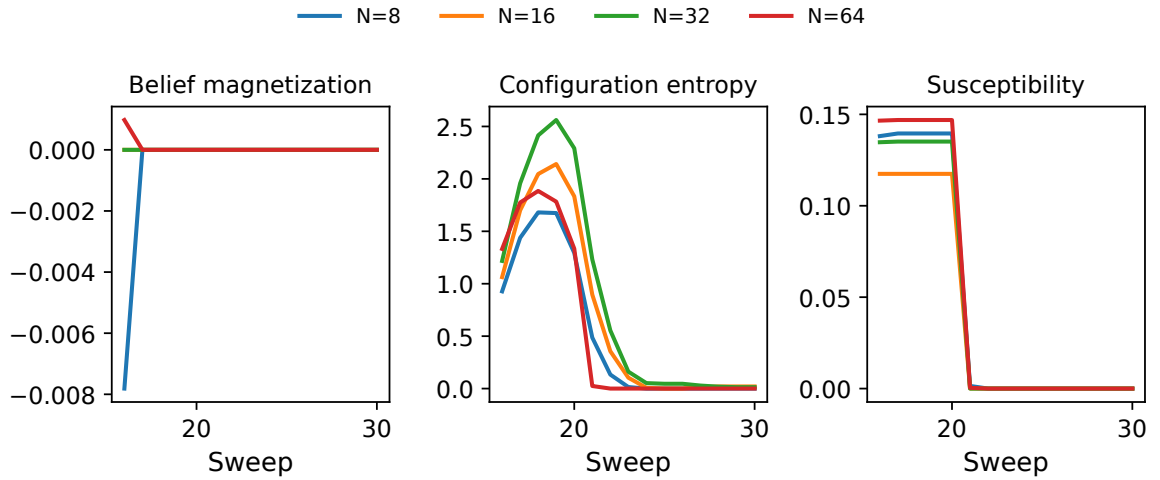


Figure 13: Direct $N = 16$ anchors in a denser inexpensive effective-model size context. These are CPU kinetic-surrogate quench responses, not direct-LLM finite-size scaling.

The autocorrelation sum is not an equilibrium correlation time, and these short, potentially nonstationary windows cannot establish critical slowing down. The Binder values are not a crossing analysis and do not establish a critical point or phase transition. Figure 14 therefore presents persistence, Binder, and occupancy together rather than as an additional confirmatory hypothesis family.

D Computational implementation and validation

An agent receives only its own persistent state, the delivered neighbor packets available at its scheduled update, and the condition specific prompt content defined in Section 2. Its response changes the scheduled instance’s recorded belief, action, commitment, memory code, signal, and typed tool consequence, the scheduler validates but does not replace those fields. Transition records include the scheduled identity, recipient, packet bytes, and prompt and output hashes. Dedicated checks confirm that perturbing one instance’s memory or decision provider leaves unrelated agents’ private state unchanged.

Every experimental arm instantiates a fresh network state from its prespecified panel seeds. Model weights and the tokenizer are shared read-only for throughput, but conversational histories, generation caches, mutable agent objects, and scientific random-number state are not carried between arms. Each generation resets its per-decision seed. The two pinned model revisions reported in Section 4.2 use Transformers 4.55.4, PyTorch 2.8.0+cu128, bitsandbytes 0.47.0, NF4 double quantization, and BF16 computation under the common decoding settings given there.

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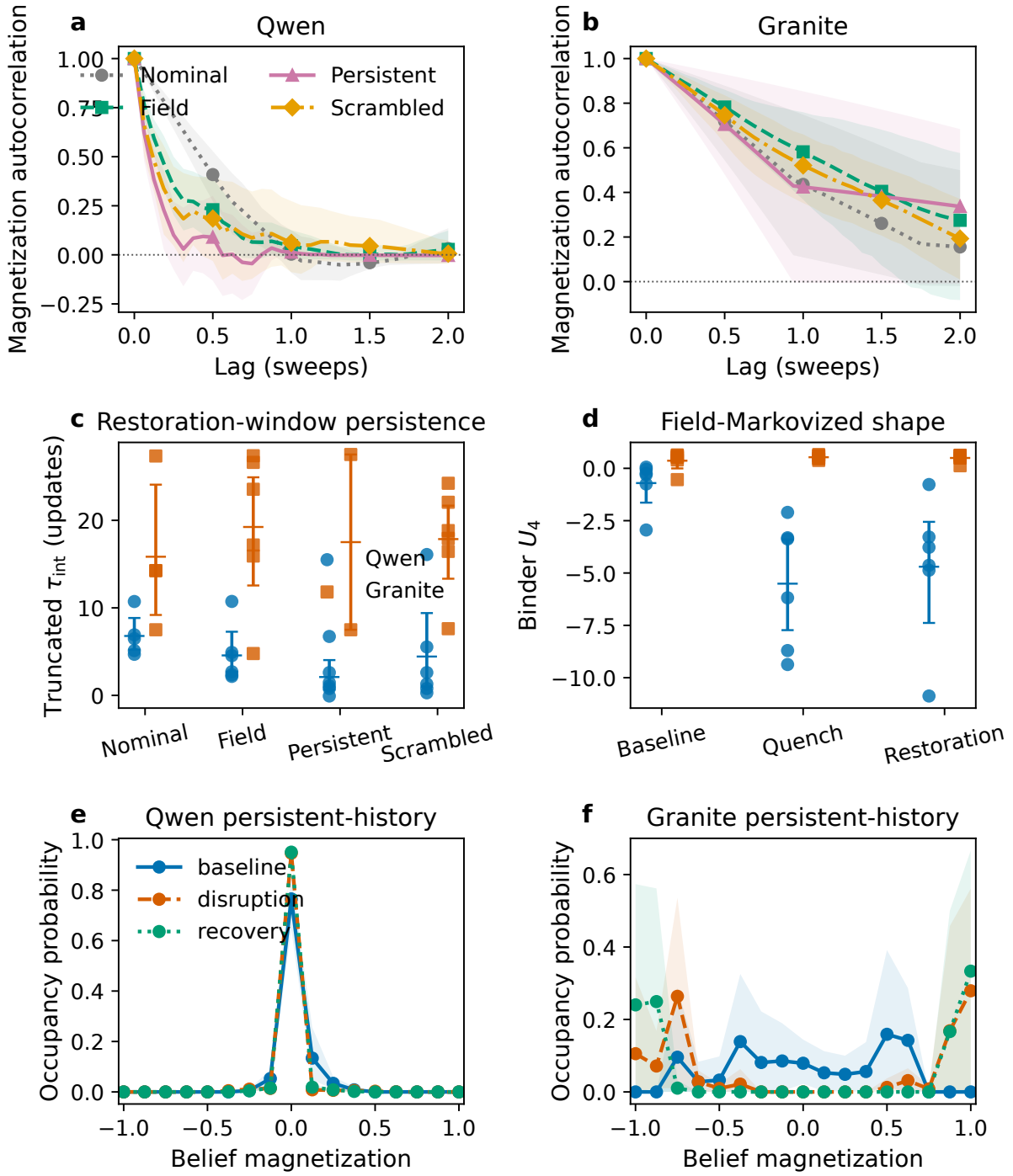


Figure 14: Finite-window persistence and order-parameter shape. Autocorrelation curves and their 95% intervals compare the nominal late window with the three field arms using complete clusters, integrated correlation sums show all cluster values at the fixed two sweep truncation. Binder cumulants are paired with empirical belief magnetization occupancies so their finite sample meaning remains visible. These diagnostics do not establish critical slowing down or a phase transition.

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